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Paper 1: QSP-Copilot: An AI-Augmented Platform for Accelerating Quantitative Systems Phar

Bibliographic Information

- **Title**: QSP-Copilot: An AI-Augmented Platform for Accelerating Quantitative Systems Pharmacology Model Development.
- **Authors**: Anuraag Saini, Ali Farnoud
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Paper Type: AI_ML

Quick Take

QSP-Copilot is an autonomous multi-agent, LLM-driven platform designed to accelerate Quantitative Systems Pharmacology (QSP) model development across project scoping, literature extraction, network construction, and ODE code generation. In two rare disease benchmark datasets (blood coagulation and Gaucher disease), the system achieved 99.1%–100.0% extraction precision and cut knowledge synthesis timelines by approximately 40%.

Executive Summary

Quantitative Systems Pharmacology (QSP) modeling is notoriously resource-intensive, with biological literature review and model structuring typically consuming 60%–70% of total project effort. Saini and Farnoud present QSP-Copilot, an end-to-end multi-agent artificial intelligence framework that automates document parsing, relation extraction, knowledge graph consolidation, interaction diagram rendering, and initial ordinary differential equation (ODE) formulation. Utilizing a multi-LLM routing strategy (GPT-4o, Claude Sonnet, and TxGemma), LangChain-orchestrated retrieval-augmented generation (RAG), and SciSpacy-based entity standardization, the platform extracts mechanistic entity-interaction pairs with high fidelity. Benchmarking against published QSP models for blood coagulation (10 papers) and Gaucher disease (9 papers) demonstrated extraction precision of 99.1% and 100.0%, with recalls of 93.8% and 86.1%, respectively. By enforcing human-in-the-loop validation checkpoints and standardizing model exports into SBML, mrgsolve, RxODE, and Python/SciPy, QSP-Copilot establishes an operational blueprint for integrating agentic generative AI into model-informed drug development (MIDD).

Scientific Context & Motivation

QSP models are increasingly leveraged to inform strategic go/no-go decisions, dose selection, and trial design in drug development. However, biological knowledge curation is fragmented and manual, creating severe bottlenecks during model scoping and structure formulation. In rare disease indications, where mechanistic pathways are sparsely documented across disparate

literature, developing a fit-for-purpose QSP model often takes months. While previous machine learning applications in pharmacometrics focused on parameter estimation, neural ODEs, or automated covariate search, fully agentic architectures capable of parsing unstructured scientific texts into executable, standards-compliant systems pharmacology models have not been realized.

⚡ Methodological Snapshot

QSP-Copilot employs a modular, agent-based architecture coordinated via Directed Acyclic Graph (DAG) task execution. The data pipeline begins with automated PDF text extraction, document chunking, and semantic embedding generation via OpenAI text-embedding-3-large, stored in a Pinecone vector database. Retrieval-Augmented Generation (RAG) is orchestrated through LangChain (v0.1.0), deploying cosine similarity thresholding (>0.7) to fetch relevant mechanistic contexts from parsed literature.

Downstream extraction and reasoning tasks leverage dynamically routed LLMs: GPT-4o for broad biomedical synthesis, Claude 3.5 Sonnet for multi-step mechanistic reasoning, and Google TxGemma for domain-specific biological entity identification. Extracted raw entity-interaction pairs undergo automated semantic deduplication and normalization using spaCy's pre-trained biomedical pipeline (en_core_sci_lg). A visualization agent compiles validated relationships into standardized pathway effect diagrams.

Finally, a code generation agent maps the finalized biological network topology into coupled ordinary differential equations (ODEs), generating ready-to-simulate code templates across multiple systems pharmacology environments (SBML, mrgsolve, RxODE, and Python SciPy solve_ivp). Strict provenance tracking is enforced, pairing every interaction node and parameter placeholder with source metadata, page numbers, and modeling assumptions.

🤖 Task Framing

The core machine learning objective is biomedical Relation Extraction (RE) and Named Entity Recognition (NER) from unstructured PDF scientific manuscripts, framed as an automated knowledge graph construction and differential equation code generation task. Solving this problem eliminates the weeks of manual literature curation and subjective diagram drawing that currently limit the responsiveness and scalability of quantitative systems pharmacology models in drug discovery pipelines.

Dataset Description

The benchmarking dataset comprises two distinct disease corpora: (1) Blood Coagulation Cascade: 10 peer-reviewed scientific articles totaling 44 pages of dense mechanistic text; (2) Gaucher Disease Type 1: 9 peer-reviewed articles totaling 39 pages. Ground truth for evaluation was established by senior pharmacometricians performing exhaustive manual extraction and comparing against previously published, validated QSP structural models (Nayak et al. for coagulation; Bultsma et al. for Gaucher disease).

Model Architecture

Multi-agent ensemble architecture using LangChain (v0.1.0) and Pinecone vector store. LLM backends include OpenAI GPT-4o / GPT-4o-mini, Anthropic Claude 3.5 Sonnet, and Google TxGemma. Embeddings generated via text-embedding-3-large (cosine similarity cutoff > 0.7). Entity extraction and standardization handled by spaCy's en_core_sci_lg model. Workflow orchestration managed as a Directed Acyclic Graph (DAG) with JSON-schema inter-agent data passing.

Training Details

No direct model fine-tuning was performed; the platform leverages zero-shot and few-shot in-context learning with domain-specific biomedical prompt templates and retrieval augmentation. Experiments tested model output variability across GPT-3.5-Turbo, GPT-4.1-nano, and GPT-4o-mini. Compute infrastructure is hosted on Amazon Web Services (AWS) with an open-source Streamlit frontend.

Evaluation Metrics

Binary classification performance on mechanistic entity pairs was evaluated using Precision, Recall, and F1-score against expert-curated ground truth. In Blood Coagulation: Precision = 99.1% (105/106), Recall = 93.8% (105/112), F1 = 96.4%. In Gaucher Disease: Precision = 100.0% (68/68), Recall = 86.1% (68/79), F1 = 92.5%. Operational efficiency was assessed by wall-clock curation time (14–17 minutes for extraction; ~40% reduction in total scoping timeline).

Deployment Caveats

Because LLMs exhibit run-to-run stochasticity and context truncation on long or complex multi-column PDFs, automated extractions cannot be ingested blindly into regulatory dossiers without human sign-off. Models should not be used as autonomous simulation engines until automated parameter estimation, structural identifiability checks, and physiological rate law specifications are fully implemented.

□ Key Findings

Across 10 blood coagulation manuscripts (44 pages), QSP-Copilot extracted 179 biological entity pairs, yielding 105 unique mechanistic interactions after automated deduplication and noise filtering in ~17 minutes (Precision: 99.1%, Recall: 93.8%, F1: 96.4%). In Gaucher disease (9 papers, 39 pages), the pipeline extracted 151 pairs, yielding 68 unique mechanistic interactions in ~14 minutes (Precision: 100.0%, Recall: 86.1%, F1: 92.5%). The platform reduced biological scoping and diagram generation turnaround from approximately 2 weeks to 6 days (~40% time savings). Generated interaction networks can be automatically exported as initial ODE structural code across SBML, mrgsolve, RxODE, and SciPy formats with full citation and provenance tracking.

Strengths & Limitations

Strengths

- First comprehensive, end-to-end multi-agent LLM framework tailored specifically to the standard 6-stage QSP workflow.
- Rigorous human-in-the-loop design requiring expert approval at critical milestones (extraction, diagram generation, ODE structure formulation).
- Robust interoperability with standard pharmacometric simulation tools (SBML, RxODE, mrgsolve, SciPy solve_ivp) compliant with MIASE guidelines.
- High factual precision (>=99.1%) achieved by grounding extractions with direct page/paragraph citations and SciSpacy NER post-processing.

Limitations (Acknowledged by Authors)

- Sub-optimal recall rates (86.1% to 93.8%), driven by LLM context window limits, token truncation, and complex multi-column/tabular PDF parsing artifacts.
- Probabilistic output variability across repeated runs of identical PDF documents using identical temperature parameters.
- Generated ODE models currently populate parameters and initial conditions with placeholder values, lacking automated parameter estimation or cellular birth/death dynamics.

Limitations (Expert Review)

- Evaluation cohort was modest (10 papers for coagulation, 9 for Gaucher), leaving scalability to broad immuno-oncology or autoimmune indication spaces (typically 30–50+ papers) unvalidated on a single end-to-end run.
- The 40% efficiency improvement is reported from a single internal retrospective benchmark rather than a randomized multi-operator productivity trial.
- Automated translation from qualitative interaction graphs into quantitative mass-action/Michaelis-Menten kinetic equations relies on generic assumptions that require substantial manual algebraic refinement.

Generalizability

Highly generalizable across biomedical literature mining and initial pathway network mapping. However, translation into fully parameterized, predictive QSP models still requires significant domain expert intervention for kinetic rate laws, initial concentrations, and clinical validation datasets.

Figures & Tables

- **Figure 1**: Resource and time allocation distribution across a standard 6-stage end-to-end QSP project execution lifecycle.
  - *Significance*: Demonstrates that biological knowledge review, scope delineation, and model structuring consume 60%–70% of total project effort, identifying the exact operational bottleneck targeted by AI automation.

- **Figure 2**: System architecture diagram of QSP-Copilot illustrating data ingestion, RAG vector indexing, multi-agent coordination, knowledge graph extraction, effect diagram rendering, and ODE code generation.
  - *Significance*: Provides the blueprint for agentic AI design in pharmacometrics, highlighting the DAG-orchestrated inter-agent communication and human validation checkpoints.
- **Figure 4**: Head-to-head network topology comparison between QSP-Copilot's automatically extracted blood coagulation interaction diagram and the benchmark peer-reviewed manual model from Nayak et al.
  - *Significance*: Visually validates that automated extraction recovers multi-level clotting factor cascades (II, V, VII, VIII, IX, X, XI, XII), regulatory proteins (PC, APC, TFPI), and fibrin mesh formation.
- **Figure 6**: Confusion matrices displaying True Positives, False Positives, and False Negatives for mechanism extraction in blood coagulation (105 TP, 1 FP, 7 FN) and Gaucher disease (68 TP, 0 FP, 11 FN).
  - *Significance*: Quantifies the machine learning performance, proving near-perfect precision while exposing residual recall gaps caused by document parsing and context window limits.

Code & Reproducibility Assessment

Publicly accessible via GitHub repository (https://github.com/QSP-Copilot/QSP-Copilot) as an open-source Streamlit application supporting basic PDF extraction and ODE generation, with cloud-native AWS enterprise deployment blueprints available.

Supplementary Materials

Supplementary Document 1 contains detailed tables (Tables S1 and S2) listing all extracted biological entity pairs, mechanistic classifications, and textual page citations for both case studies, alongside high-resolution pathway effect diagrams (Figures S1 and S2).

Future Directions

1. Integration of automated parameter estimation workflows connecting extracted pathways directly to experimental databases (ChEMBL, UniProt, PK-DB).
2. Automated kinetic rate-law selection based on biological context (e.g., distinguishing Hill-type cooperative binding from simple mass-action kinetics).
3. Autonomous reconciliation and triangulation against curated pathway ontologies (Reactome, STRING, Pathway Commons) to proactively address false-negative recall gaps.

Expert Commentary

QSP-Copilot is an important milestone in modern pharmacometrics, marking the transition from narrow AI tasks (e.g., parameter optimization, neural ODEs) to agentic workflow orchestration. The authors demonstrate sound engineering judgment: recognizing that LLM hallucinations are intolerable in regulatory-grade systems pharmacology, they bounded the agents by strict

page-level citation tracing, SciSpacy NER normalization, and mandatory human approval gates. The near-zero false-positive rate (99.1%–100%) proves that RAG-grounded LLMs can effectively parse dense pharmacological literature. Nonetheless, pharmacometricians should recognize its current ceiling: the tool delivers an automated conceptual effect diagram and a baseline ODE topology with placeholder parameters. The hardest intellectual work of QSP—identifying structural unidentifiability, tuning physiological feedback gains, and generating virtual populations—remains firmly in human hands.

Bottom Line

QSP-Copilot demonstrates that agentic LLM workflows can reliably slash literature curation and interaction-diagram drafting from weeks to days with >99% precision, provided domain pharmacometricians remain in the loop to supervise parameterization and kinetic rate assumptions.

Fact-check corrections

Figures

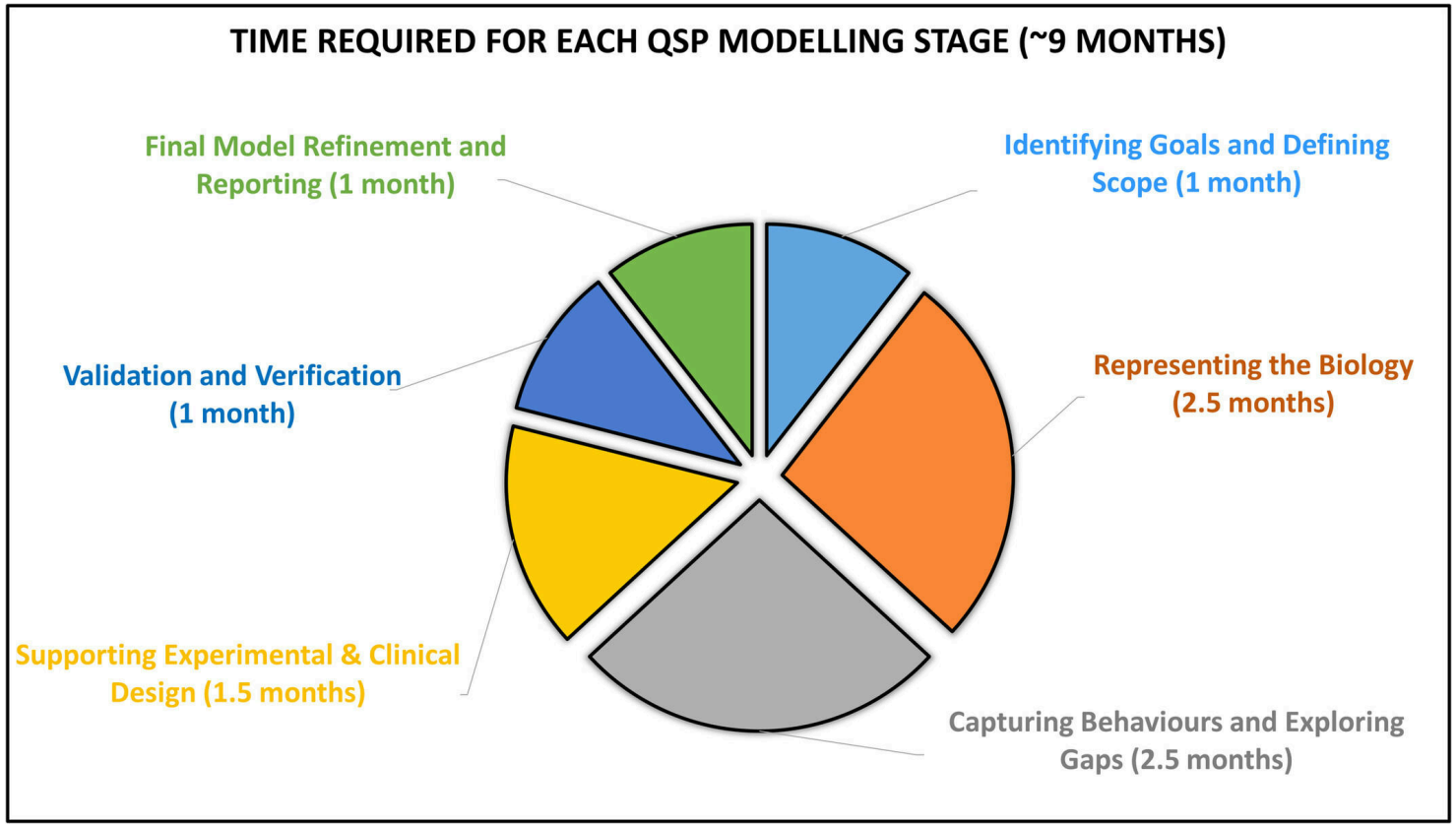


Figure 1: Time required across different stages in a typical mid-term end-to-end QSP project execution.

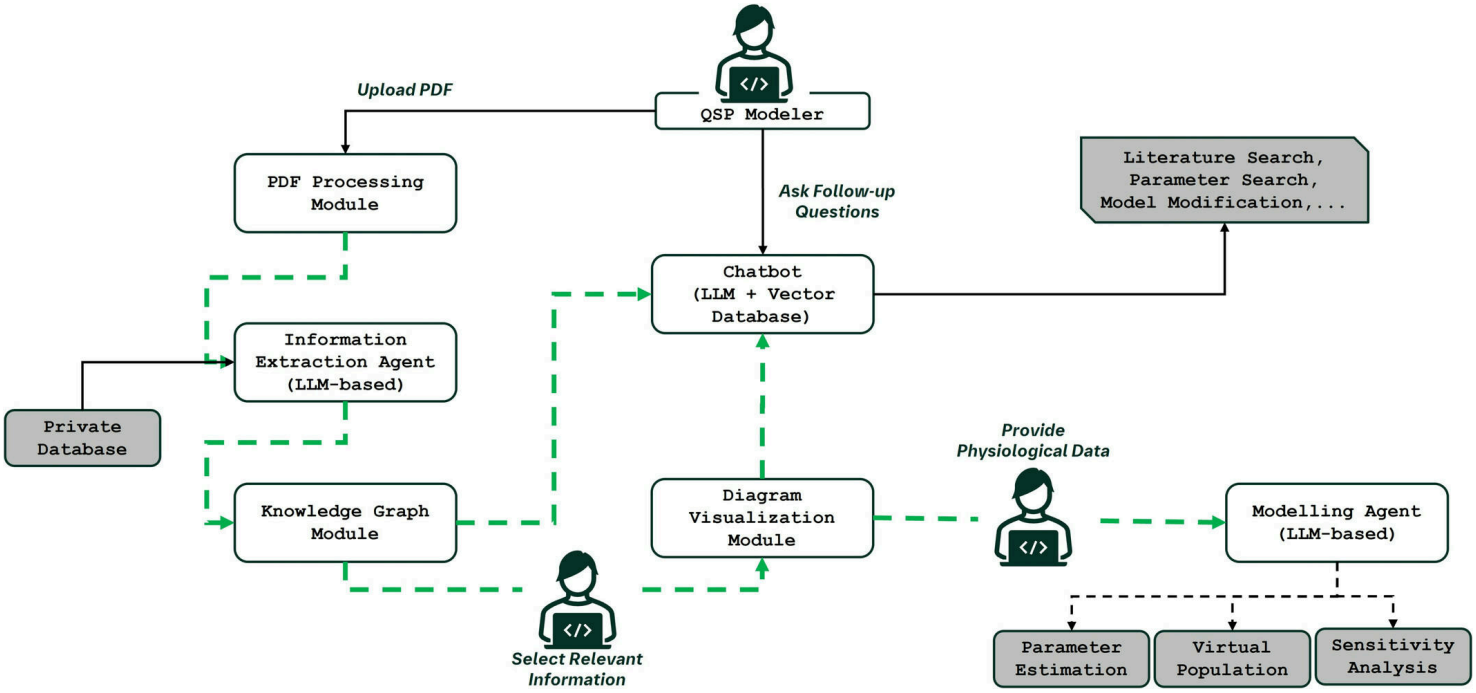


Figure 2: System architecture of QSP-Copilot.

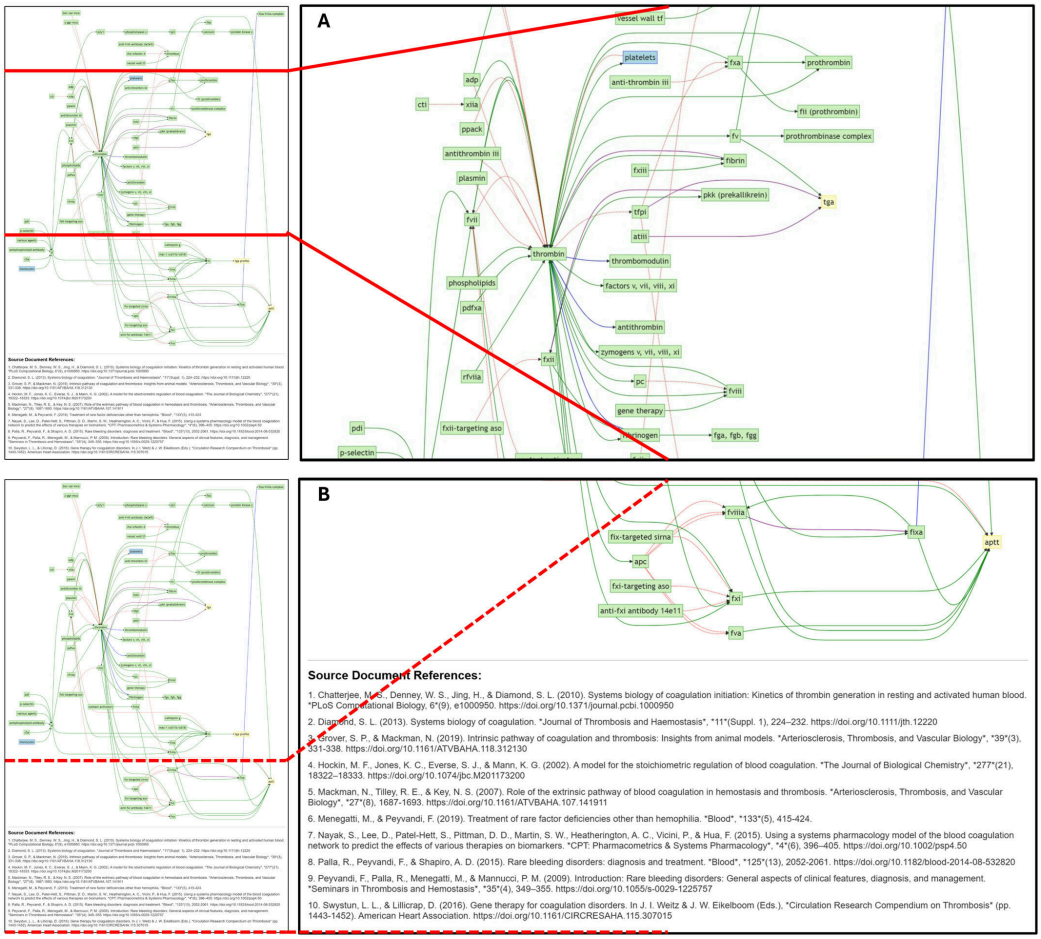
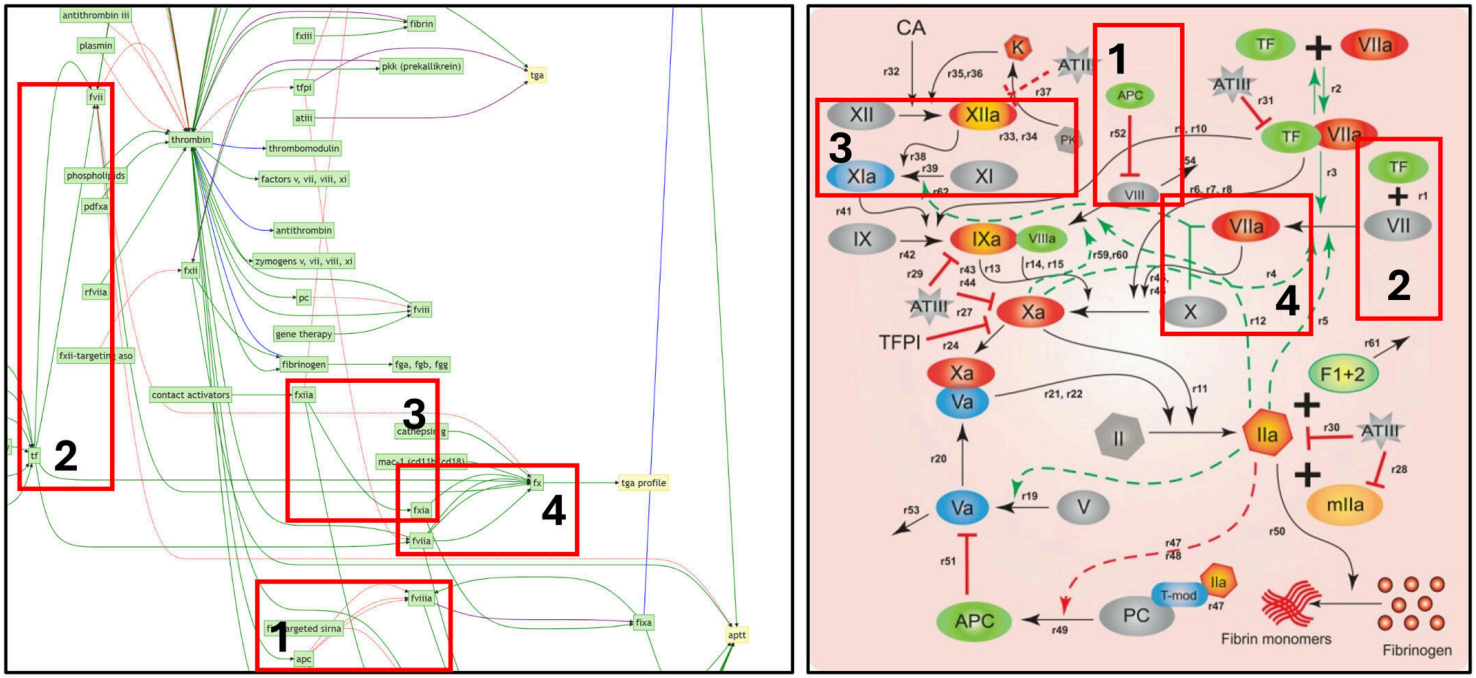


Figure 3: (A) Blood coagulation regulation network diagram and (B) List of 10 scientific reference sources used for generating the diagram 37,39-46.



Interaction diagram generated using QSP-copilot

Figure taken from Nayak et al. 2015

Figure 4: Interactions between the biological entities extracted using QSP-Copilot (left) versus interaction diagram reported in Nayak et al. (right) 37.

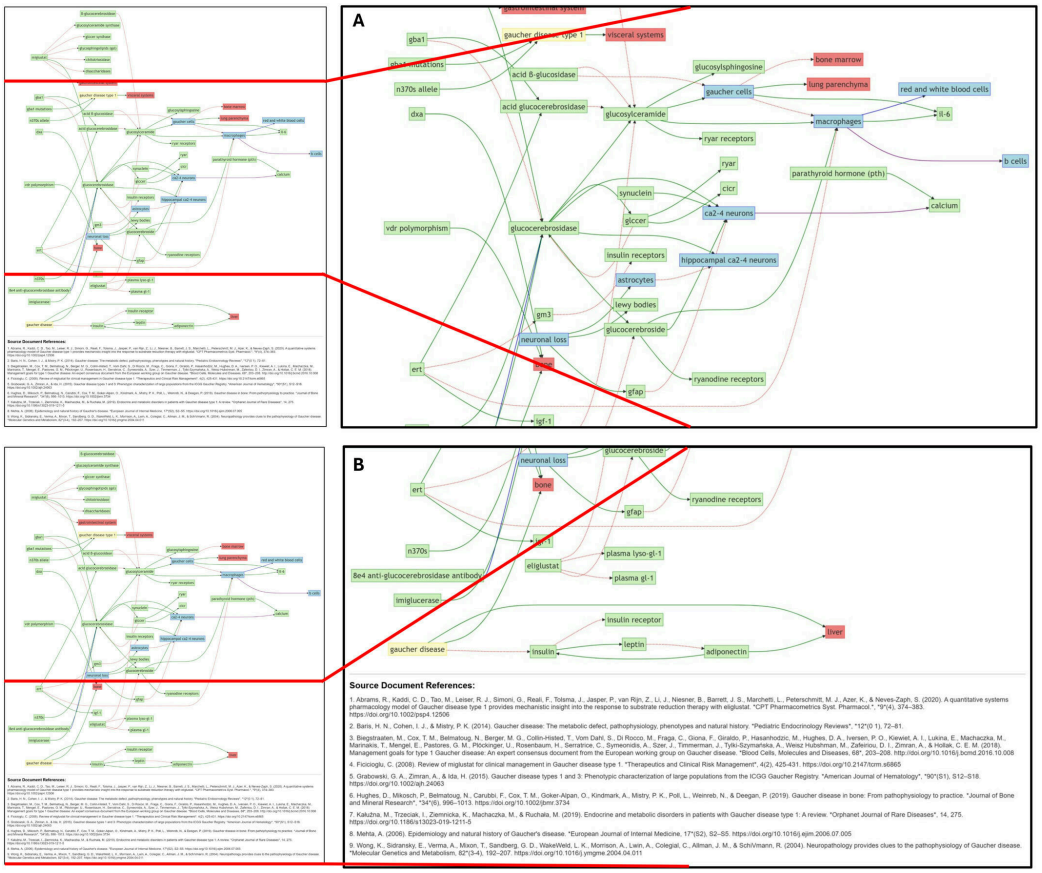


Figure 5: (A) Gaucher Disease biological entity interaction diagram and (B) List of 9 scientific reference sources used for generating the diagram 38,47-54.

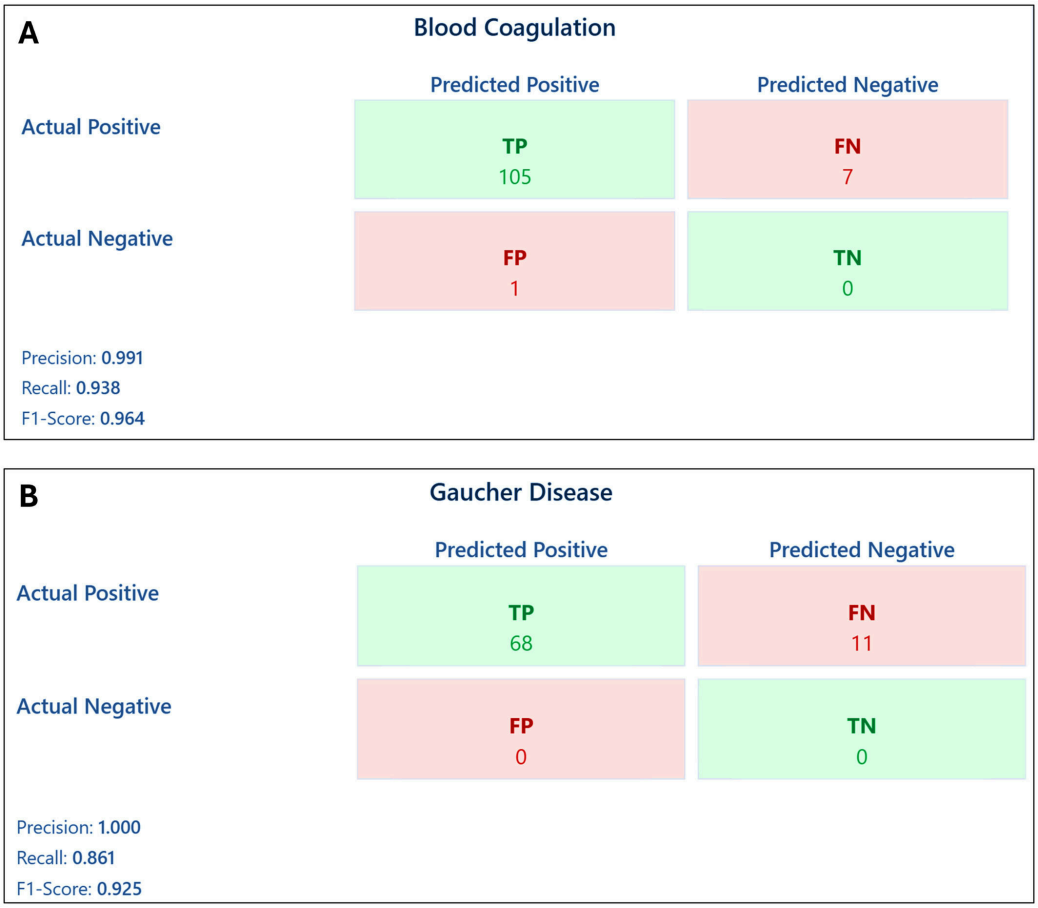


Figure 6: Confusion matrices for biological mechanism extraction performance from literature in (A) blood coagulation and (B) Gaucher disease.

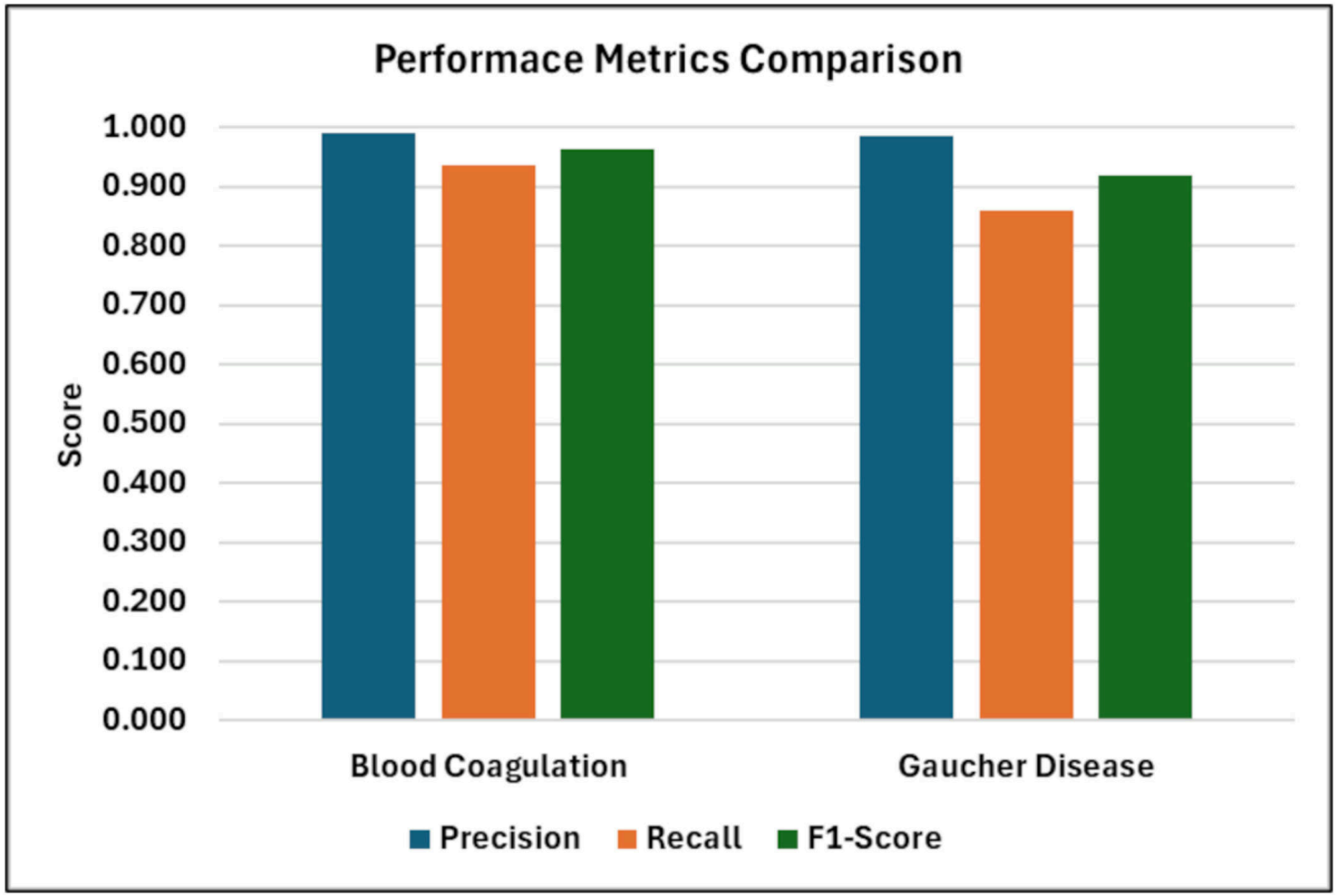


Figure 7: Performance metrics comparison between blood coagulation and Gaucher disease extraction.